

Continuous Distillation

Pre-Lab Plan
Unit Operations II Lab 3
Date: 2/20/20

Group 2, Thursday, PM section

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Commented [PTH1]: 32/35

1. Experimental Objective

In this experiment, a binary distillation column will be examined. It will be separating ethanol and water by operating at steady state with a continuous feed, a top product reflux ratio and a bottom product. The steady state distillation of the binary mixture experiment will be performed along with varying the feed position.

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mention that you'll be using McCabe-Thiele analysis for the column efficiency

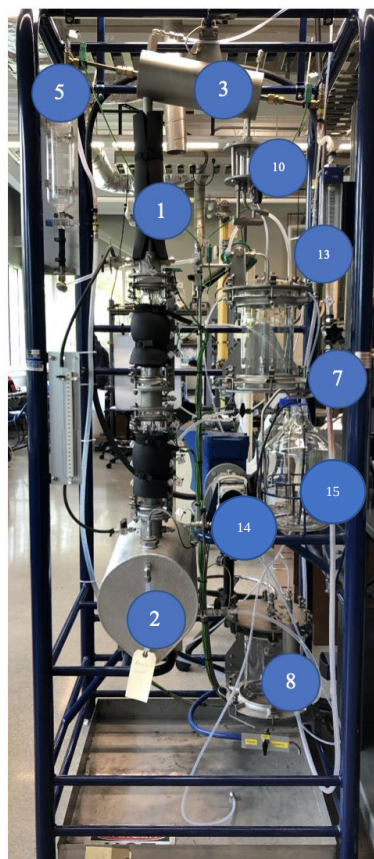
2. Experimental

2.1 Apparatus

The apparatus setup for this experiment includes a distillation column, pump for the feed, a condenser, and a reboiler. The following figure (Figure 1) and table (Table 1) summarize the setup for this lab.

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missing the Armfield control and computer

Front view:



Side view:

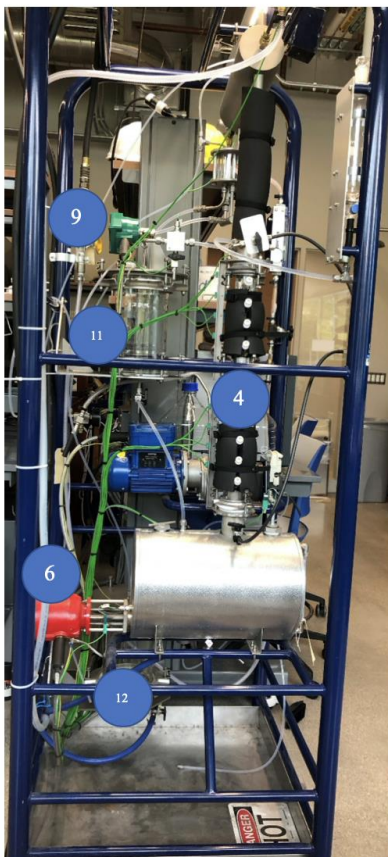


Figure 1: Apparatus setup for Continuous Distillation experiment

Table 1: Continuous Distillation experiment apparatus and equipment

Number	Component Name	Description
1	Column	Separates liquid mixtures based on differences in volatility
2	Reboiler	Provides heat to the bottom of the distillation column
3	Condenser	Changes vapor to liquid in the distillation column
4	Trays	Perforated plates in the distillation column
5	Manometer	Measures the pressure of a column
6	Heater	Gives heat to the reboiler
7	Valves	Controls fluid flow through a pipe
8	Heat exchanger	Transfers heat from one stream to another
9	Reflux valve	Controls the reflux rate in the distillation column
10	Decanter	Separates mixtures of liquids and solids through suspension
11	Top product receiver	Receives top product (methanol)
12	Bottom product receiver	Receives bottom product (water)
13	Cooling water flowmeter	Cools vapor in condenser to liquid
14	Peristaltic feed pump	Pumps feed from jug into column
15	Feed holding jug	Holds feed solution

2.2 Materials and Supplies

The following table summarizes the material needed for this experiment and a description of their purpose in the lab.

Table 2: Material List with Descriptions

Index	Material/Supply Name	Description
1	Ethanol	One component in the mixture
2	Water	One component in the mixture
3	Large graduated cylinder	Used to make ethanol and water solution
4	Measuring cylinder	Used to collect samples during timed collection
5	Timer	Used to time collection duration
6	Bottle	Used for waste collection during lab
7	Refractometer	Used to measure refractive index
8	Test tubes	Used to hold samples while cooling
9	Computer	Runs UOP Distillation program
10	Armfield console	Controls column parameters

Commented [PTH4]: 4.5/5

Commented [PTH5]: computer belongs to the apparatus section

Commented [PTH6]: also belongs to the apparatus section

2.3 Experimental Plan

Experiment A

1. Produce a calibration graph of the actual flowrate against the position of the variable speed dial on the control console
2. This can be done by taking measurements of flow at 10 settings over the full range
3. Disconnect the feed tubing to the distillation column
4. Ensure the feed tank with the pump suction pipe inserted has sufficient water for the calibration
5. Produce the graph

Experiment B

Set Up

1. Make up 10 L of a mixture of 10wt% ethanol/water and 3 L of a mixture of 30wt% ethanol/water
2. Set equipment to operate at total reflux by switching pff the reflux ratio timer on the console (ensure all valves are closed)
3. Load the 10 L solution through the filler cap on top of the reboiler using the funnel/blue hose
4. Fill one of the feed bottle with 3 L of the 30wt% solution
5. Connect the blue cap to the bottle and connect the black tubing from the feed bottle to the feed pump

Record

- 3 sets of top sample refractive index (one after 500 mL fed, 3 more samples 5 min apart)
- 3 sets of bottom sample refractive index (one after 500 mL fed, 3 more samples 5 min apart)

Procedure

1. Follow detailed setup procedures in manual
2. Open valve V5 to set the cooling water flow rate to 3 L/min
3. Turn on the reboiler element and set it to 1.5 kW
4. Once vapor has traveled up the column and distillate begins to collect in the receiver, turn the reboiler power down to 0.75 kW
5. Set the reflux ratio to 5:1, then switch on the reflux valve.
6. Wait 15-30 min for the column to stabilize
7. Open valve V1 to let the bottom product exit
8. Turn on the pump to feed 30-35 mL/min to the column
9. After feeding ~500 mL to the column, take a sample of the top and bottom products
10. Take two more samples of the top and bottom products 5 minutes apart

Experiment C

1. Turn on reboiler power to 0.75kW
2. Let column stabilize
3. Repeat procedures in Experiment B, but with a different feed position

Commented [PTH7]: 9/10
missing dependent and independent variables.

Project Safety Evaluation

Commented [PTH8]: 10/10

Project Title: Continuous Distillation

Date: 2/20/20

	Yes/No	
Potential electrical hazards?	Yes	If yes, identify conditions: Control panel is an electrical hazard
Potential mechanical hazards?	Yes	If yes, identify conditions: Moving valves could pinch
Potential pressure hazards?	No	If yes, identify conditions:
Potential temperature hazards?	Yes	If yes, identify conditions: Column is hot, the samples collect are hot – USE CAUTION
Hazardous/toxic chemicals involved?	Yes	If yes, hazards involved: Ethanol – denatured ethanol contains solvents dangerous for consumption
Gloves required?	Yes	If yes, specify type: Latex gloves when preparing solution and performing experiment
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: Water and Ethanol
Process Parameters	Max Expected	List location and possible hazards and precautions
Temperature (°C)	~90°C	Column temperature, reboiler, samples collected
Pressure (psig)	ambient	none
Safety Controls	yes/no	If yes, Provide description
	yes	Proper heating of column required, see instructor or TA once solution filled in column prior to starting, see operation manual for column, Fuse to shut off column Reboiler level indicator if too low will not heat

Commented [PTH9]: near water source

I have read relevant background material for the Unit Operations Laboratory entitled: “Continuous Distillation” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

<u>Jake Proctor</u>	<u>2/20/2020</u>
signature of team member	date
<u>Courteney Fabbri</u>	<u>2/20/2020</u>
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<u>Milos Markovic</u>	<u>2/20/2020</u>
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